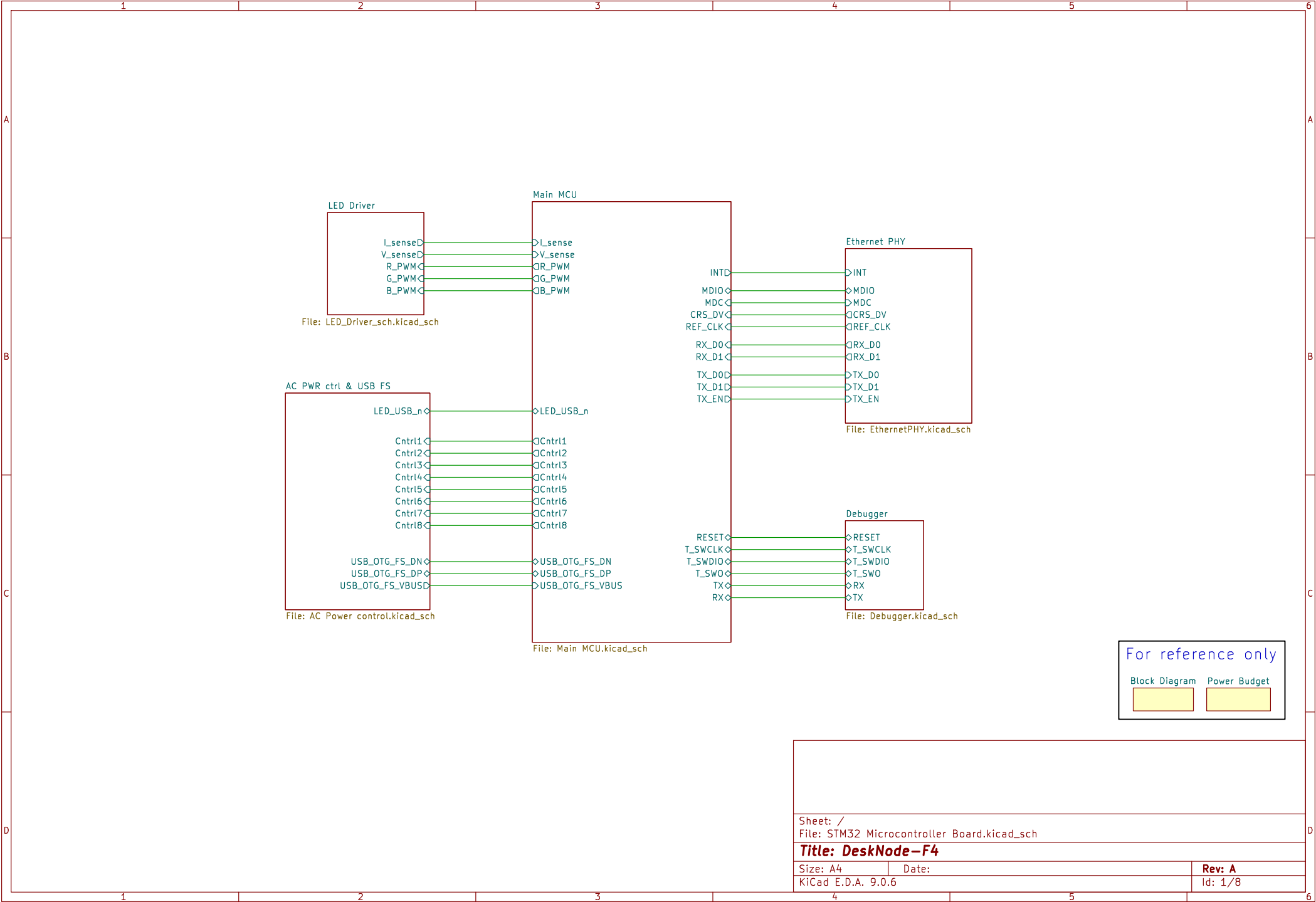
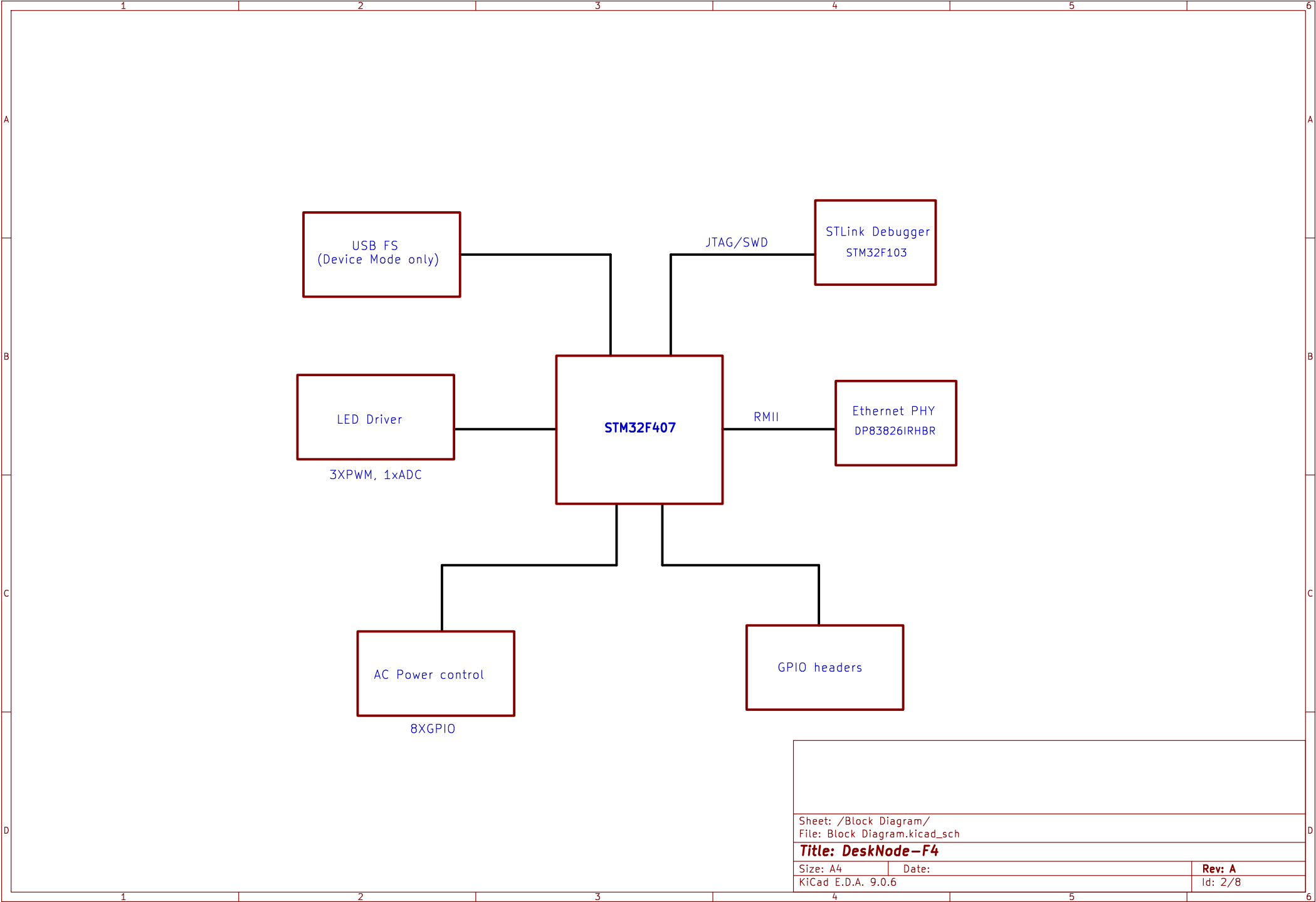
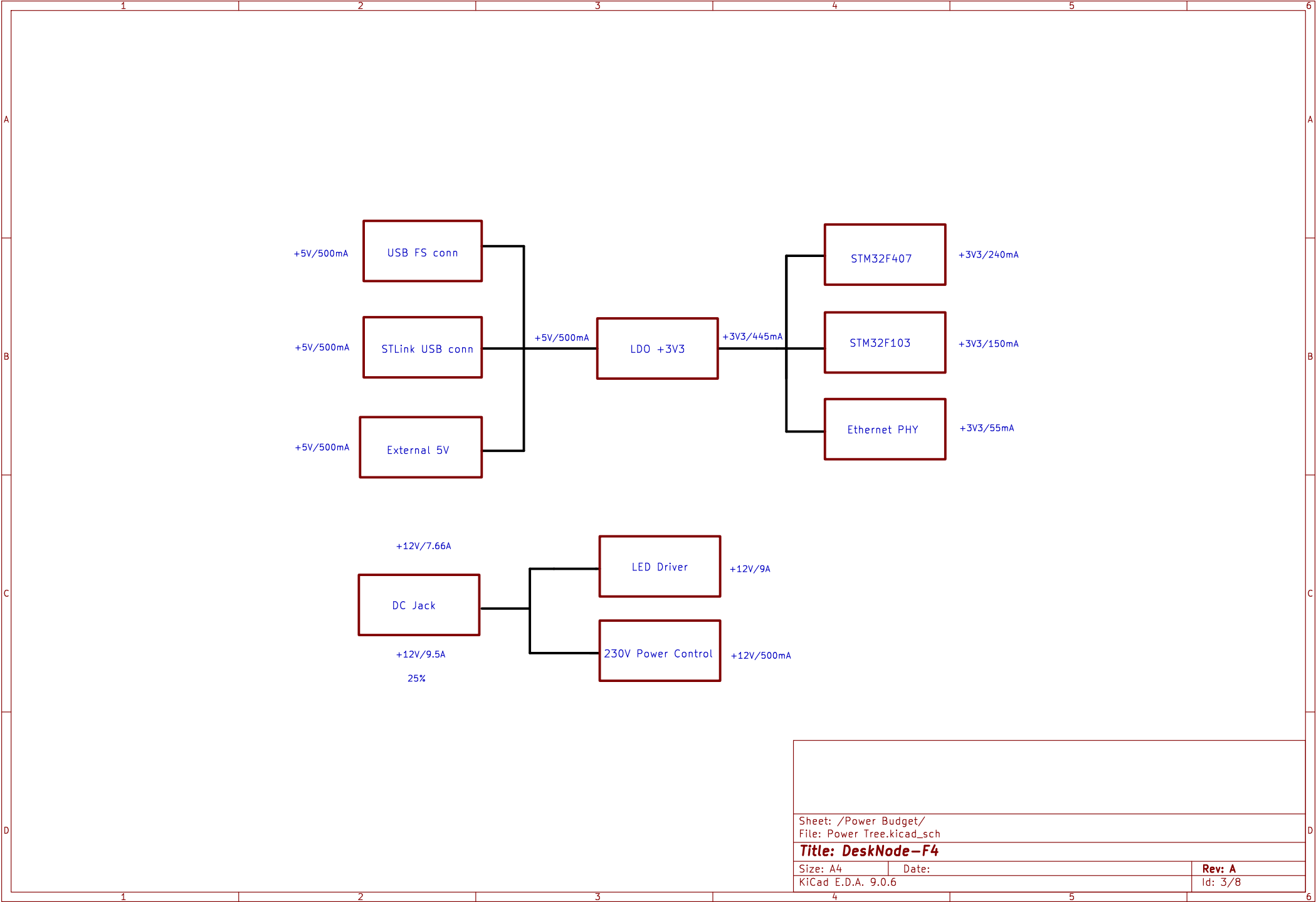


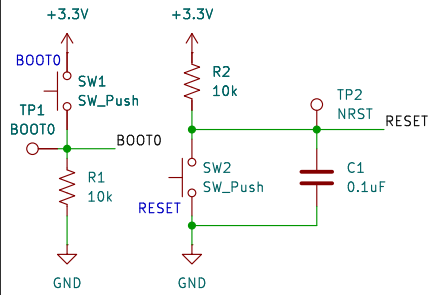




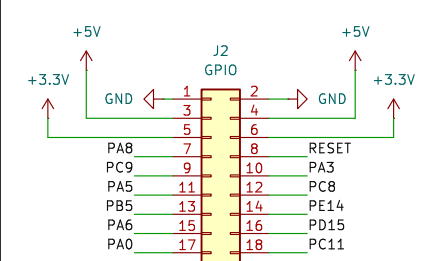
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Title: DeskNode-F4		
Size: A4	Date:	Rev: A
KiCad E.D.A. 9.0.6	Id: 2/8	



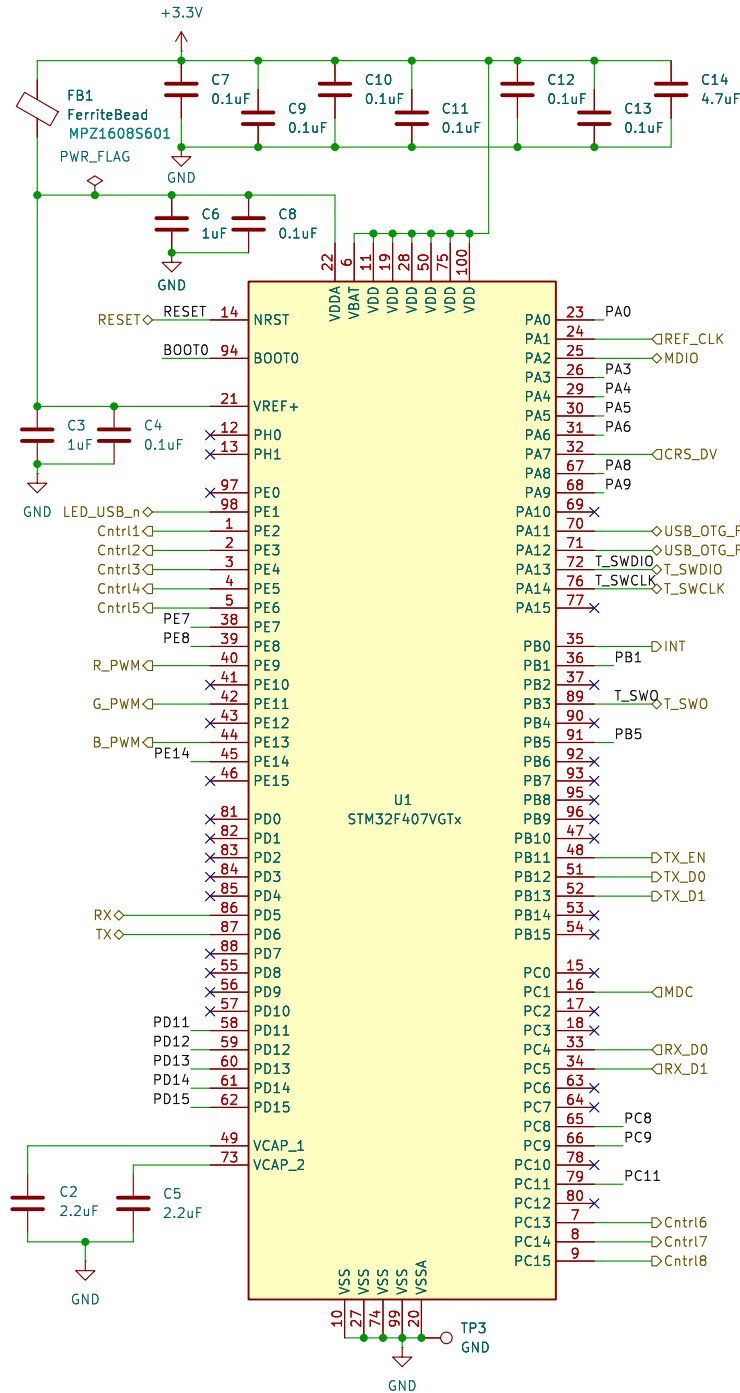
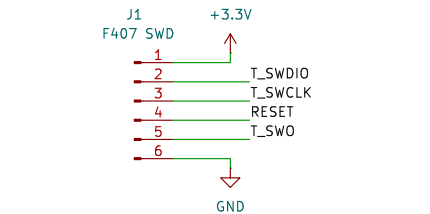
STM32F407 RESET & BOOT0



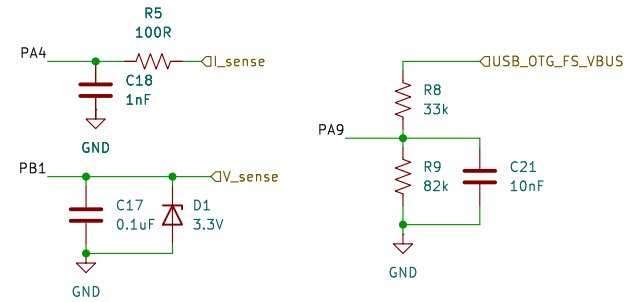
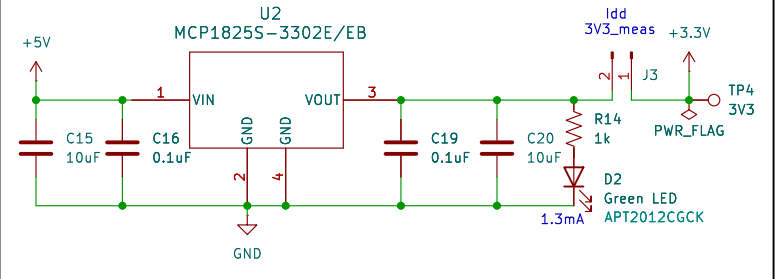
GPIO Extension



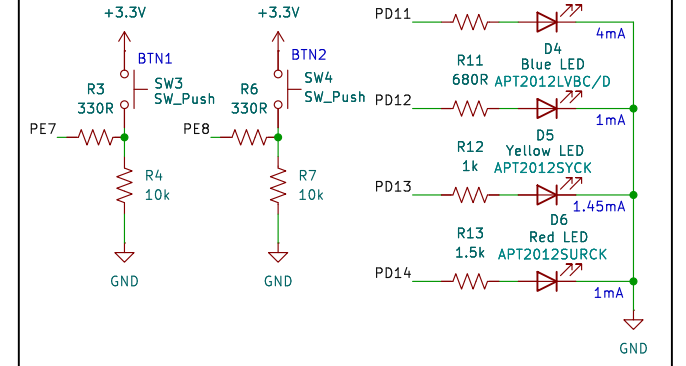
STM32F407 SWD



3V3 Power Supply



Peripherals



Sheet: /Main MCU/
File: Main MCU.kicad_sch

Title: DeskNode-F4

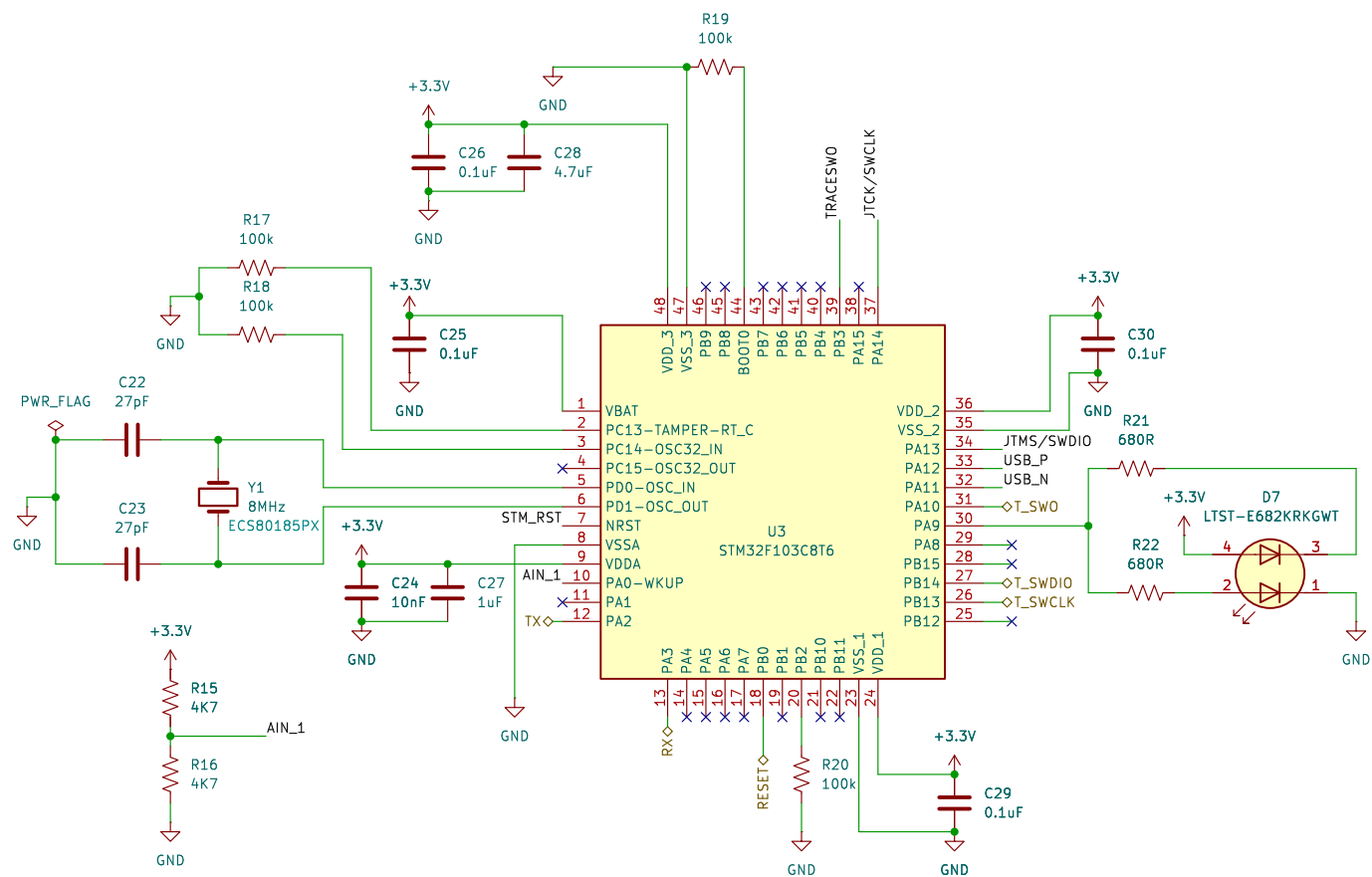
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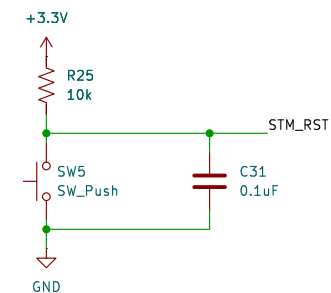
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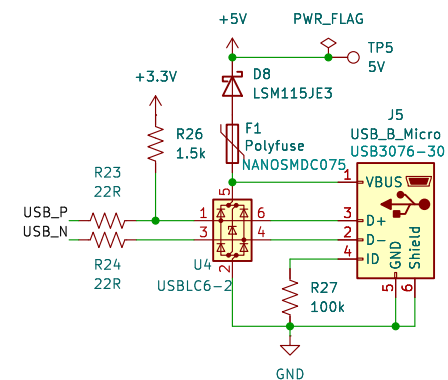
Debugger/Programmer



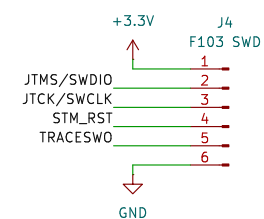
Reset STM32F103



USB2.0 for Program/Debug



STM32F103 SWD



Sheet: /Debugger/
File: Debugger.kicad_sch

Title: *DeskNode-F4*

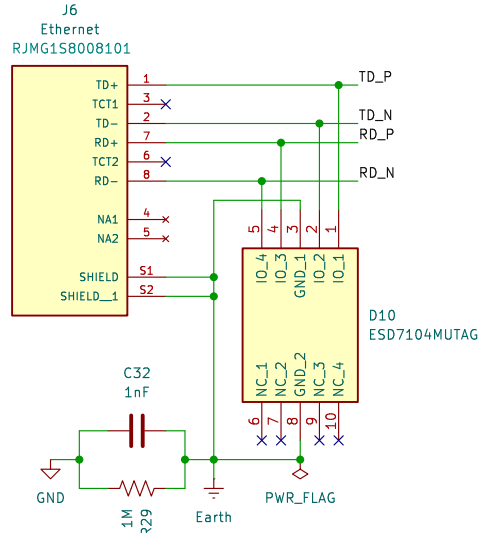
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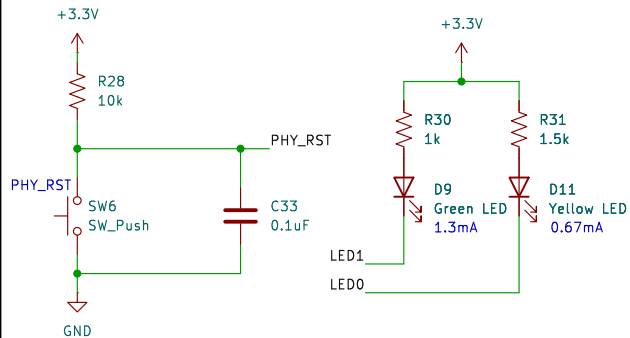
Rev: A

Id: 5/8

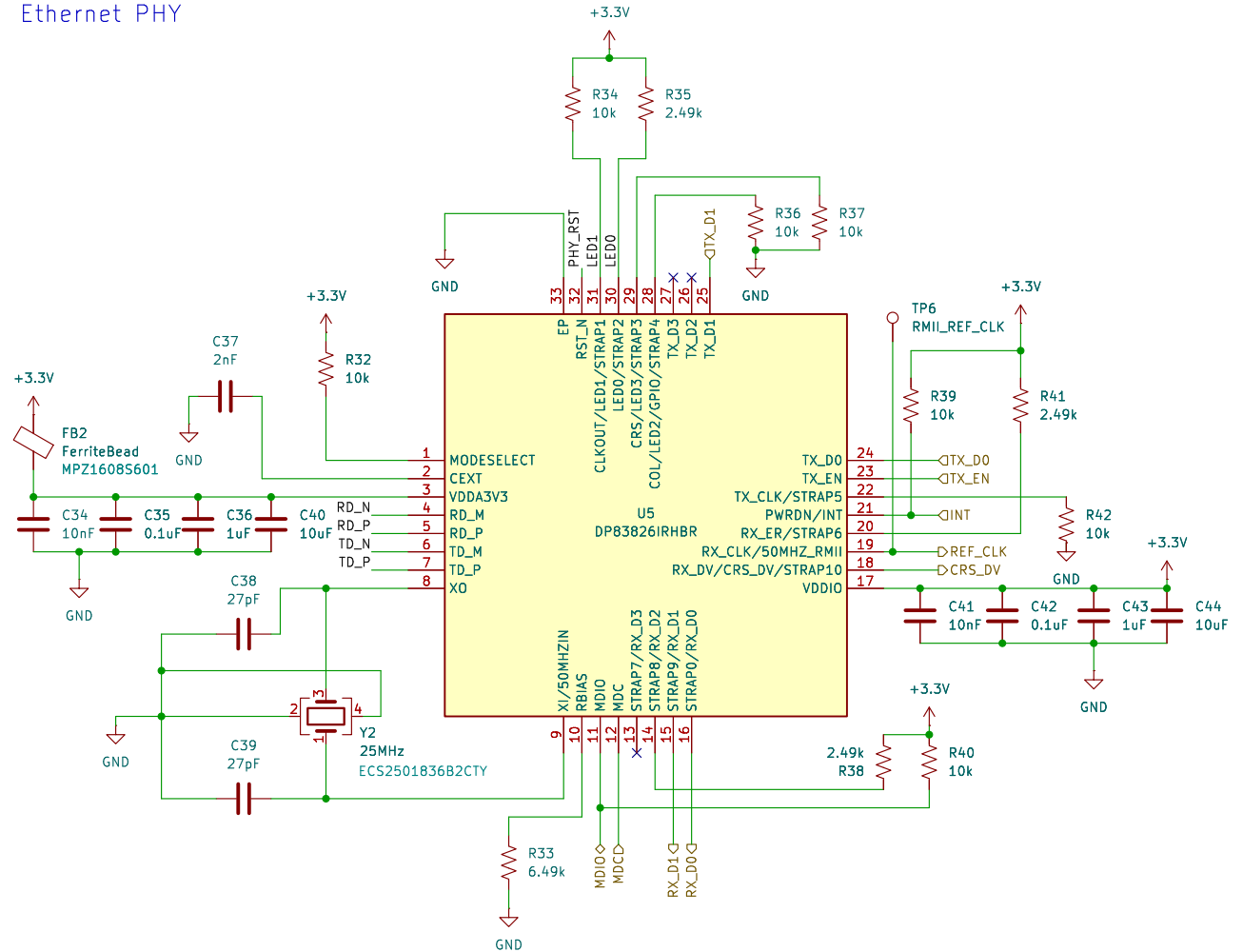
RJ45 Connector



ETH Reset and Status LEDs



Ethernet PHY



PHY Bootstrap table

Pin Name	Strap	Pin No	Strap Fn	Default	Resistor	Final State
RX_D0	Strap0	16	Auto-negotiation en/dis	0	-	0 - enabled
LED1	Strap1	31	Old Nibble en/dis	1	10k	1 - enabled
LED0	Strap2	30	PHY_ADD0	0	2.49k	1
LED3	Strap3	29	PHY_ADD1	0	10k	0
LED2	Strap4	28	PHY_ADD2	0	10k	0
TX_CLK	Strap5	22	RMII Leader/Follower	0	10k	0 - Leader
RX_ER	Strap6	20	Pin 31 is CLKOUT 25MHz/LED1	0	2.49k	1 - LED1
RX_D3	Strap7	13	Pin 18 is CRS_DV/RX_DV	0	-	0 - CRS_DV
RX_D2	Strap8	14	MII/RMII	0	2.49k	1 - RMII
RX_D1	Strap9	15	auto MDIX en/dis	0	-	0 - enabled
RX_DV	Strap10	18	MDIX/MDI (only applicable if auto-MDIX is dis)	0	-	0 - NA

Sheet: /Ethernet PHY/
File: EthernetPHY.kicad_sch

Title: DeskNode-F4

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12V RGB LED Conn

J8
12V RGB LED

1 R_LED
2 G_LED
3 B_LED
4

B4PS-VH(LF)(SN)

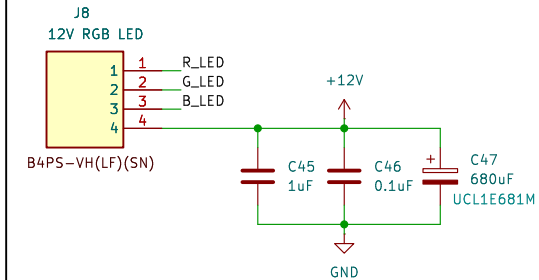
+12V

C45 1uF

C46 0.1uF

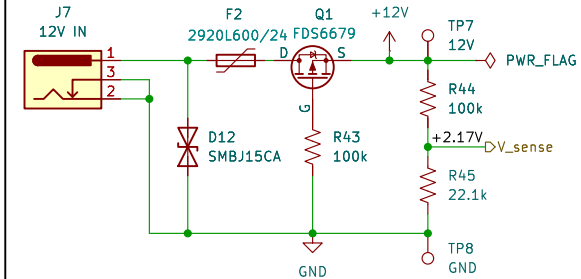
C47 680uF UCL1E681M

GND



12V LED Power

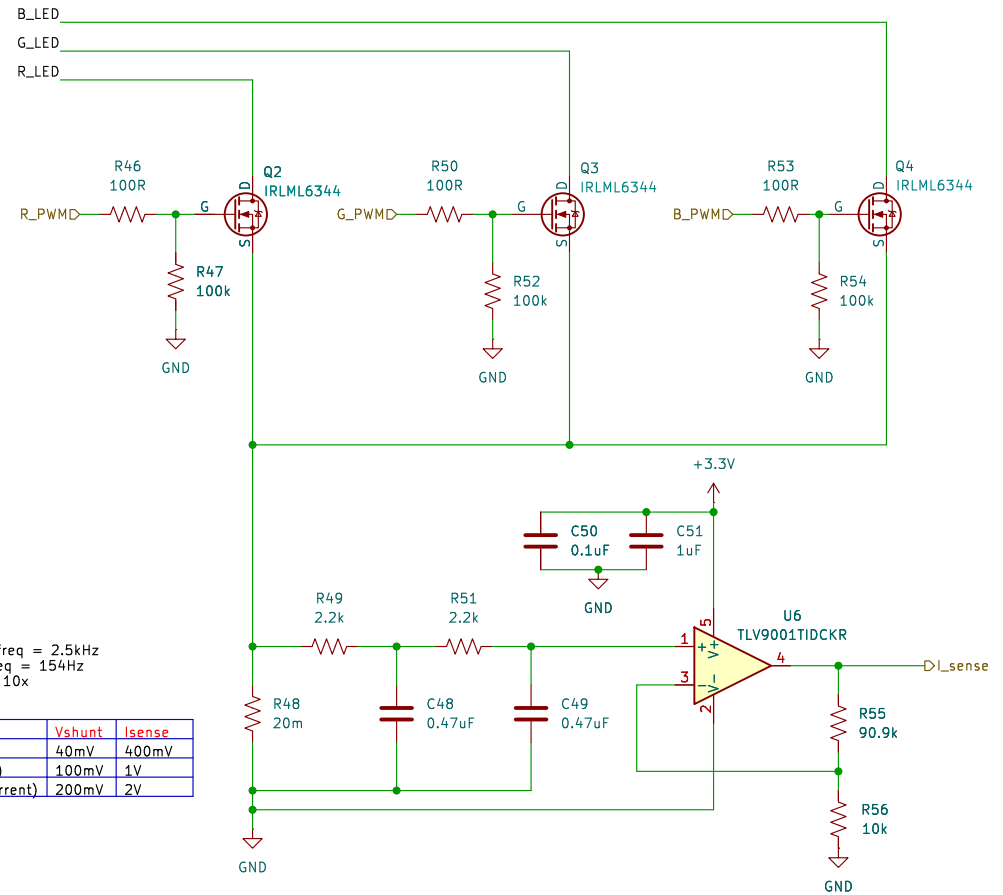
The diagram illustrates a 12V LED Power circuit. It features a 12V IN source connected to a fuse F2 (2920L600/24) and a diode D12 (SMBJ15CA). The MOSFET Q1 (FDS6679) is used for switching. The gate of Q1 is connected to a +12V source through a 100k resistor R44. The drain of Q1 is connected to a +2.17V source through a 100k resistor R43. The source of Q1 is connected to GND. The +2.17V source is also connected to a 22.1k resistor R45, which is connected to GND. The output is labeled TP7 12V and PWR_FLAG.



LED Driver with current sensing

LED switching freq = 2.5kHz
Filter Cutoff freq = 154Hz
Opamp Gain = 10x

Ishunt	Vshunt	Isense
2A	40mV	400mV
5A (Max Limit)	100mV	1V
10A (Fault Current)	200mV	2V



LED switching freq = 2.5kHz
Filter Cutoff freq = 154Hz
Opamp Gain = 10x

Ishunt	Vshunt	Isense
2A	40mV	400mV
5A (Max Limit)	100mV	1V
10A (Fault Current)	200mV	2V

Title: *DeskNode-F4*

Title: DeskNode-F4

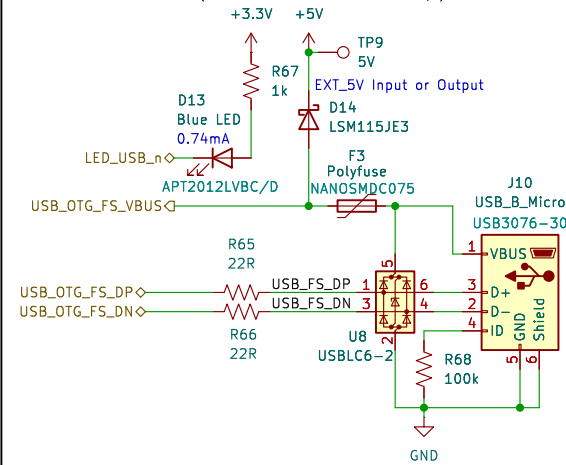
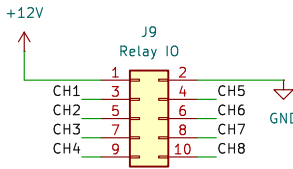
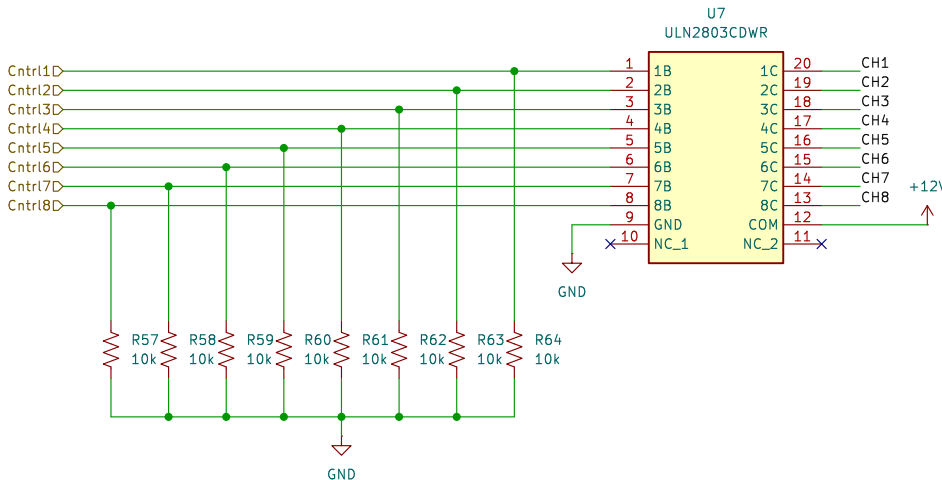
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Date:

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AC Power Control



- H1
MountingHole
- H2
MountingHole
- H3
MountingHole
- H4
MountingHole

Sheet: /AC PWR ctrl & USB FS/
File: AC Power control.kicad_sch

Title: DeskNode-F4

Size: A4

Date:

KiCad E.D.A. 9.0.6

Rev: A

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